

Poset-Stratified Hybrid State Spaces

Dan Guralnik*

Department of Mathematics,
Ohio University, Athens, OH

Abstract

Poset-stratified spaces with reset relations will be considered as an alternative to the “graph-of-spaces” approach to specifying a hybrid dynamical system.

Generalizing the setting of hybrid differential inclusions (HDI) from manifolds, a hybrid state space in the new formalism is a (nice enough) topological space X —not necessarily a manifold—equipped with a jump relation Δ . The relation Δ is used to produce a weaker (coarser) topology on X , resulting in a non-Hausdorff space X_Δ , with the property that paths in X_Δ admit a gamut of discontinuity types regulated by Δ , from jump discontinuities in X , even to essential ones, without the need for an explicit model of hybrid time tailored to each individual path.

The traditional graph structure of reset maps between modes is replaced by a poset stratification of X_Δ , a continuous mapping $q : X_\Delta \rightarrow \Gamma$, where $(\Gamma, \leq)$ is a poset endowed with its Alexandrov topology. The advantage of this description is its capability of explicitly presenting the same hybrid state space X_Δ in a variety of discretizations Γ , offering the opportunity to compare such discretizations within the common framework provided by a single category, and generalizing the recent construction of the category of charts by Moeller and Ames.

I will introduce the basic tenets of this theory, consider examples of its deployment in several settings, such as hybrid-linear systems, and provide an overview of the applications we have in mind.

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